

BROUGHTON ISLAND, NU, Canada

WMO: 710960

Lat: 67.536N

Lon: 63.789W

Elev: 1917

StdP: 13.71

Time Zone: -5.00 (NAE)

Period: 82-18

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	-32.1	-28.8	-38.5	0.6	-32.3	-36.9	0.7	-30.9	46.0	-4.5	41.0	-10.0	7.0	0	1.061

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.2	57.4	49.4	54.1	46.6	51.3	44.4	48.5	53.9	45.2	50.6	43.1	48.4	7.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
45.0	47.4	51.2	41.1	40.8	46.9	38.8	37.3	45.0	20.1	54.2	18.2	50.6	17.2	48.6	59.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 36.9	31.5	26.7	-32.2	60.8	4.4	3.5	-35.4	63.3	-38.0	65.3	-40.5	67.3	-43.8	69.8		
(5)			-33.9	50.9	3.0	3.7	-36.1	53.5	-37.8	55.7	-39.5	57.8	-41.8	60.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	13.0	-11.8	-13.5	-8.7	2.9	18.0	33.0	42.3	39.6	29.0	19.6	7.7	-4.6		
	DBStd	21.26	9.82	10.28	10.94	10.18	8.52	7.03	7.18	6.91	5.70	6.33	9.09	10.33		
	HDD50	13551	1917	1779	1820	1412	991	510	258	330	630	943	1268	1693		
	HDD65	18998	2382	2199	2285	1862	1456	959	704	787	1080	1408	1718	2158		
	CDD50	27	0	0	0	0	0	1	19	7	0	0	0	0		
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0		
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0		
	WSAvg	9.8	11.2	10.5	10.0	9.2	9.2	9.1	8.2	8.8	9.7	10.0	10.7	10.6		
	PrecAvg	11.4	0.4	0.4	0.2	0.6	1.3	1.1	0.9	1.1	1.6	2.0	1.4	0.4		
	PrecMax	18.1	2.9	1.5	0.9	2.0	5.3	4.4	2.2	2.7	4.3	6.5	3.1	1.5		
(15) Precipitation	PrecMin	7.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.3	0.4	0.2	0.1	0.0		
	PrecStd	2.9	0.6	0.4	0.2	0.4	1.1	1.1	0.7	0.7	0.9	1.2	0.8	0.4		
	0.4%	DB	23.2	23.4	26.8	32.5	45.1	55.9	62.2	59.8	48.8	40.7	32.0	26.1		
	0.4%	MCWB	22.0	22.7	25.1	29.0	39.5	43.7	58.7	48.4	43.1	40.2	31.8	25.4		
	2%	DB	13.2	14.4	17.8	26.4	38.2	50.4	58.9	55.7	44.0	34.2	26.1	19.6		
(23) Mean Coincident Wet Bulb Temperatures	2%	MCWB	12.4	13.5	16.8	25.6	34.9	43.8	51.7	45.4	39.7	32.2	25.4	18.7		
	5%	DB	6.2	5.4	11.6	22.0	33.4	46.6	56.0	52.7	40.7	30.2	22.9	14.9		
	5%	MCWB	5.5	4.7	10.7	21.2	32.1	40.8	49.1	44.0	37.4	28.9	22.0	14.1		
	10%	DB	1.5	-0.8	6.7	17.3	29.5	43.3	53.1	49.9	37.3	27.5	19.9	9.4		
	10%	MCWB	1.0	-1.4	5.8	16.6	28.9	38.5	45.8	43.3	35.2	26.7	19.1	8.7		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	13.5	24.5	22.1	27.4	36.9	45.7	57.4	49.6	47.0	40.6	30.7	18.7		
	0.4%	MCDB	14.4	24.8	23.6	27.9	38.2	50.4	59.6	57.0	52.8	42.8	30.8	19.2		
	2%	WB	6.6	9.1	12.0	23.6	33.8	43.4	51.1	44.7	40.7	31.4	23.0	14.8		
	2%	MCDB	7.0	9.6	12.6	24.3	36.0	48.3	54.5	50.7	43.8	32.3	23.5	15.6		
	5%	WB	2.4	1.9	6.4	18.5	30.6	41.1	47.7	43.0	36.4	28.3	20.5	9.5		
(31) Mean Daily Temperature Range	5%	MCDB	2.9	2.5	7.0	19.2	31.7	44.9	53.0	48.0	38.6	28.9	21.5	10.2		
	10%	WB	-2.0	-3.7	0.5	14.0	27.6	38.7	44.8	41.1	33.9	25.3	16.4	4.2		
	10%	MCDB	-1.6	-3.1	1.1	14.8	28.3	42.7	50.4	45.3	35.6	26.1	17.1	4.8		
	5% DB	MDBR	7.5	7.8	8.8	8.6	7.1	7.3	8.2	7.4	5.0	5.4	6.3	7.2		
	5% DB	MCDBR	11.5	13.4	12.8	13.2	11.5	11.1	10.3	10.2	9.6	8.6	9.1	10.8		
(36) Clear-Sky Solar Irradiance	5% DB	MCWBR	10.8	11.5	15.1	14.2	8.7	7.5	6.6	8.0	6.6	9.1	9.2	10.1		
	5% WB	MCDBR	9.5	11.6	14.6	12.0	9.1	9.7	9.2	9.8	7.8	9.0	9.2	10.1		
	5% WB	MCWBR	9.5	11.5	14.6	12.3	8.3	8.6	8.1	6.9	6.3	9.2	9.3	10.1		
	taub		0.146	0.192	0.233	0.277	0.307	0.302	0.313	0.311	0.272	0.227	0.153	N/A		
	taud		1.861	2.096	2.176	2.077	2.072	2.250	2.400	2.469	2.463	2.203	1.928	N/A		
(42) All-Sky Solar Radiation	Ebn at Noon		63	215	265	277	279	285	279	268	252	191	56	0		
	Edh at Noon		3	15	25	37	42	36	30	25	19	13	3	0		
(44) All-Sky Solar Radiation	RadAvg		16	202	756	1525	2100	2176	1753	1147	649	241	33	0		
	RadStd		1	11	31	69	75	149	127	70	44	17	3	0		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+1.22	+2.70	N/A	+0.92	N/A	N/A	-454	-470	+4	N/A
(47) Regional (5 neighbors)		+1.12	+2.63	+3.20	N/A	N/A	N/A	-392	-400	N/A	N/A

Nomenclature: See separate page